

Thursday 2-5

Characterizing Shelf Life of Novel DLP 3D Printing Resin

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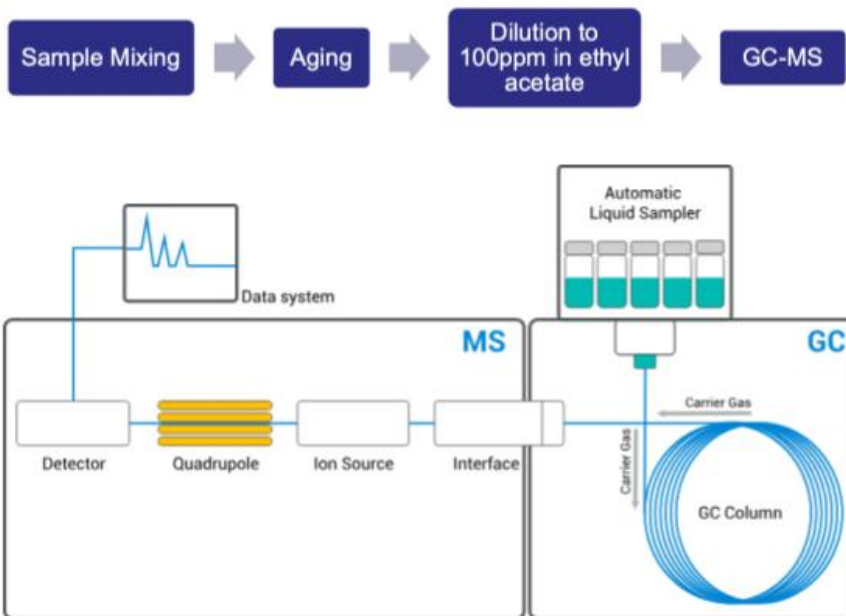
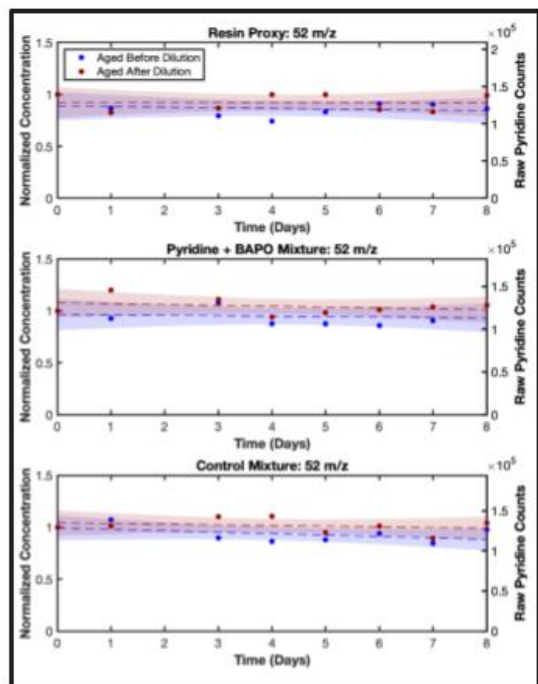
2.671 Measurement and Instrumentation

Thursday PM

Prof. Lachner

Visual Abstract

Characterizing Shelf Life of DLP 3D Printing Resin



Abstract

Understanding the shelf life of resins used for DLP (Digital Light Processing) 3D printing is critical to producing quality parts and reducing waste. Researchers at MIT have developed a resin capable of producing parts with soluble supports, and the stability of this resin is not yet known. This study investigated the stability of this novel resin through GC-MS measurements of pyridine, an important component of the resin which is hypothesized to decay in the resin over time. Results of GC-MS measurements on 3 different resin proxies with two different preparation methods showed that pyridine levels did not change significantly over a 9-day aging period. The results indicate that pyridine is most likely not the chemical responsible for the shelf-life of the resin. However other chemical interactions within the resin may contribute to its degradation over time and warrant further investigation.

Keywords: Shelf-life, Gas Chromatography Mass Spectrometry, GC-MS, Pyridine, Stability, 3D printing

1. Introduction

Digital Light Processing (DLP) 3D printing is an additive manufacturing method widely used in the dental, medical, and jewelry industries due to its ability to produce parts with fine features, smooth surface finishes, and intricate designs that may be difficult or impossible to make through more traditional manufacturing methods. DLP 3D printing forms each layer of a part by projecting the shape of the layer onto a photopolymer resin, hardening it and fusing it to previous layers. This method is significantly faster than stereolithography (SLA) printing, which uses laser vector scanning to trace a layer line by line, and it is significantly more precise and higher quality than Fused Deposition Modeling (FDM) printing, which creates parts with high roughness and low precision. For these reasons, DLP 3D printing is an extremely powerful manufacturing tool, and has already been proven to be invaluable in various industry applications.

Recently, researchers in the Mechanosynthesis Group at MIT have developed a method for DLP 3D printing parts with soluble supports. This is made possible with a novel dual-monomer resin, which allows two separate polymer networks to be formed within the same print. This opens the door to producing extremely complicated parts, allowing users to make parts with supports that would traditionally be very difficult or impossible to remove manually. This has huge implications for industry applications of DLP technology, allowing for faster production, increasing design freedom, and eliminating unwanted surface features left from manual support removal.

This resin is very new, so there are still a handful of unknowns around its usage. One of these is its shelf life. In order to make it practical to use in industry, it is important to quantify how long the resin is able to last under normal storage conditions (dark, relatively dry, room temperature). With this information, users of this new resin will be able to maintain good print quality, limit waste, and more effectively plan out resin consumption levels and print schedules.

It has been observed that older resins produce prints with undesired and excessive epoxide polymerization in the supports, which as a result become less soluble. The chemical responsible for temporal control of the epoxide network is pyridine, which acts as a weak base to slow down cationic polymerization. It is hypothesized that pyridine interacts with both the photo acid generator and the free radical generator in the resin. Thus, in order to determine shelf life, the concentration of pyridine was measured over time. This was done through Gas Chromatography Mass Spectrometry (GC-MS), which has the capability of measuring the relative abundance of any chemical of interest within the resin over time, with a sensitivity high enough to distinguish chemicals with a mass difference of less than 1 dalton.

2. Background

This resin contains two base monomers: an acrylate monomer and an epoxide monomer [1]. It contains several other active ingredients which enable controlled polymerization of each polymer system, both spatially and temporally. One of these ingredients is pyridine (C_5H_5N), a 6-member ring. Pyridine acts as a weak base, so varying its concentration allows temporal control of the epoxide network polymerization. This is because the epoxide within the resin

undergoes cationic polymerization, which is initiated with a weak acid. Thus, it is expected that pyridine may react with the photoacid generator in the resin [2], leading to unwanted byproducts and depleting the functional components. In addition, the resin contains Phenylbis(2,4,6-trimethylbenzoyl)phosphine oxide (BAPO), which acts as a photoinitiator for free radical polymerization of the acrylate network. Light exposure causes BAPO to produce free radicals, a process which likely continues to some extent even when the resin is placed back in the dark.

According to a 1998 study [3], pyridine is capable of radical degradation through the pathways shown in Figure 1 below. The study investigated radical pyridine degradation under high temperature conditions rather than interactions with UV-activated radical initiators, but it is plausible that a radical initiator like BAPO may have a similar effect on pyridine and its subsequent degradation.

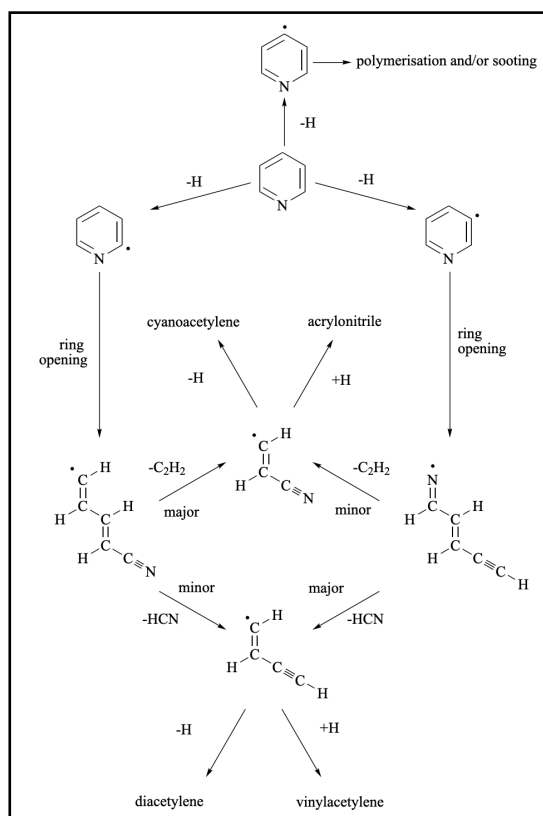


Figure 1: One anticipated pyridine reaction and resulting byproducts [3]

Because there are two potential mechanisms through which pyridine may decay in the resin, and because the resin's mode of failure implies pyridine depletion, it is worthwhile to investigate the effects of time on pyridine concentrations.

Much performance-based variation of the pyridine concentration was done during the resin development process. As such, there exists a certain known threshold at which pyridine levels are too low (around 0.0345 weight percent, a 12% decrease from the optimal concentration). Thus,

the shelf life of the resin will be defined as the time it takes for pyridine levels to reach this minimum acceptable concentration.

There does not yet seem to exist a standardized approach for determining the shelf life of photo activated resins. A 2018 study [4] on the shelf life of PFA, a thermosetting resin, performed accelerated aging on resin samples through various heat treatments. The study characterized the shelf life by measuring the resin's change in viscosity for each heat treatment, and fitting appropriate exponential models to the system. Though a similar approach can be taken for determining the shelf life of the DLP 3D printing resin, a true aging method was selected instead of accelerated aging through temperature or light treatments because it is more representative of the actual system.

Measuring the resin's characteristics over time (as the study above did) is a very good way to gauge its shelf life. However, this method does not yield information on the underlying mechanisms which cause the decrease in resin quality. For this reason, pyridine was directly measured, allowing information to be gained on the mechanism of decomposition as well as the shelf life of the resin. The instrument used for measuring pyridine levels is GC-MS. The particular machine (from MIT's Department of Chemistry Instrumentation Facility) used for this measurement requires that the sample be dissolved in a GC-MS friendly volatile solvent at a concentration of at most 100 ppm. If the resin itself were diluted to this extent, the pyridine levels would be too low to read. Instead, the monomer component of the resin was replaced with ethyl acetate (a solvent for GC-MS which does not interact with any components in the resin), and then diluted further in ethyl acetate until the appropriate concentration was reached.

Because pyridine is in very low concentrations at the time of measurement, it is important to use selected ion monitoring (SIM) mode on the machine, which targets certain molecular weights at a higher sensitivity than a broad scan. The GC-MS instrument used for these measurements employs electron ionization (as opposed to chemical ionization) in order to ionize the samples for measurement, and it's a fairly destructive method. As a result, it fragments pyridine into separate molecules with different mass to charge ratios (units m/z). Measuring more than one fragment can provide a more wholistic understand of the system and how pyridine levels are changing.

3. Experimental Design

GC-MS is a highly sensitive instrument, and is well suited to taking accurate measurements of small concentrations of chemicals. However, this also makes it a very delicate instrument, and using it to take measurements on a solution which contains significant levels of monomer is ill advised as it runs the risk creating unwanted "gunk" within the instrument. For this reason, all measurements of pyridine levels were taken on resin proxy mixtures, in which all monomer was replaced with the chosen GC-MS solvent ethyl acetate.

The first resin proxy prepared was a control sample. It contained pyridine only (in the same concentration as in the actual resin), and all other components including monomer were replaced with their weight in ethyl acetate.

The second resin proxy contained only Pyridine and BAPO (the suspected reactant causing pyridine to decay) in their true-to-resin concentrations. As with the first proxy, all other components were replaced with their weight in ethyl acetate.

The third resin proxy contained all non-monomer components of the resin in their normal concentrations, and as with the samples above, all monomer was replaced with its weight in ethyl acetate. This proxy was designed to be as representative of the true resin system as possible.

Each of the 3 proxy resins were subjected to a 9-day aging process. Before aging, each of the 3 proxies were separated into two groups. The first group was diluted for GC-MS (1000x dilution in ethyl acetate in order to achieve 100 ppm of the present chemicals) before beginning the aging process. The second group was aged in true-to-resin concentrations, and diluted after being aged.

The aging process itself took place in a dark drawer in a room-temperature environment — a fairly standard method of storing DLP 3D-printing resins. All proxy resins (pre-diluted and true-to-resin concentrations) were stored together. Over the course of 9 days, a small amount of each proxy was removed from the drawer each day, diluted for GC-MS (if not pre-diluted), and then transferred to a dark freezer. This aging process operates under the assumption that the freezer is a stable environment which would halt any degradation reactions. The process starts on day 0, which represents a “fresh” resin: samples were moved to the freezer immediately after the resin proxies were prepared. Each following day, at roughly the same time of day, samples were moved to the freezer. The end of the aging process resulted in 9 samples (representing ages of 0-8 days) per proxy per dilution method. Samples representing day 2 were spilled, and so GC-MS measurements were not able to be taken for that age.

At the end of the aging process, all samples were transferred to GC-MS vials and measured on the same day. Because GC-MS produces slightly different signals depending on what was run through the instrument previously as well as environmental factors, all samples must be run through the instrument at once and on the same day. This is also why samples were transferred to the freezer after aging instead of simply being measured day of.

3.1 GC-MS Measurements

The GC-MS instrument used is a single-quadrupole nominal mass Agilent 5977B mass spectrometer detector attached to a 7890B gas chromatograph with an autosampler. It is capable of both electron ionization (EI) and chemical ionization (CI). The method used for collecting this data was EI.

One fairly important practice for taking GC-MS measurements that was not used for these measurements is employing an internal standard. An internal standard for GC-MS is a chemical that is added to all samples at a known, precise concentration. This chemical should be similar in volatility and mass to the measured chemical of interest. Including an internal standard can allow one to control for any error that occurred during dilution, as well as allow one to make statements about absolute concentrations of the chemical of interest (rather than just counts) [5]. Unfortunately, an internal standard was not used for these measurements as it did not arrive in

time. However, the parameter of interest is a time constant with which to determine a shelf life, and this can be found directly from the measured counts of pyridine despite increased uncertainty from dilution.

In order to measure pyridine levels with GC-MS, two SIM measurements were run per sample: one on 52 m/z (fragmented pyridine ion), and one on 79 m/z (unfragmented pyridine ion). For both of these m/z values, pyridine elutes at roughly 3.25 minutes. Thus, the counts of pyridine in a certain sample correspond to the area of the 3.25 minute peak of the SIM chromatogram for the ion of interest. An example of a SIM peak for pyridine is shown in Figure 2 below. It is also worth noting that by no means is it expected or common for the counts of each ion to align — pyridine fragments unevenly so it is normal for counts of one ion to be significantly different from counts of the other ion for the same sample.

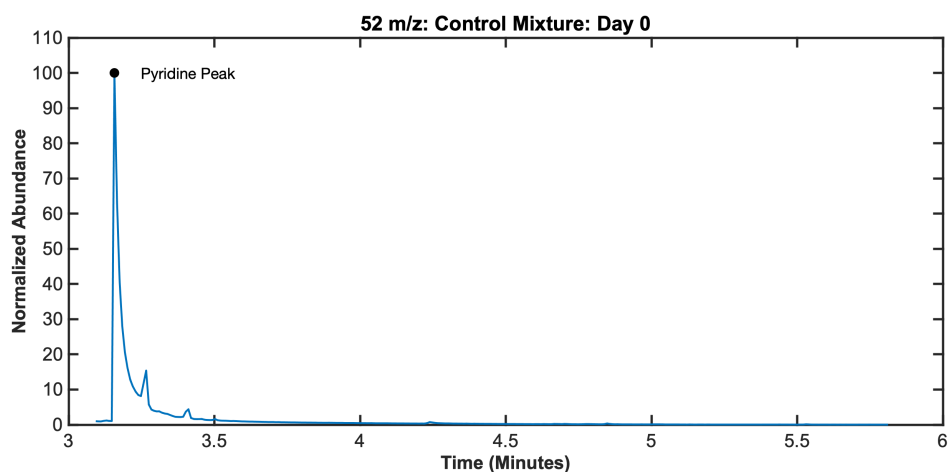


Figure 2: Normalized SIM chromatogram of 52 m/z for the control mixture aged 0 days. The non-normalized peak is integrated over time using Agilent GC-MS analysis software.

4. Results and Discussion

Remarkably, there was not a significant decreasing trend in pyridine over time for any of the tested samples. This can be seen in Figure 3 and Tables 1 and 2, which show linear fits to the data and their slopes. Both for 52 m/z and 79 m/z and for both dilution methods, pyridine levels were roughly constant over time. Additionally, there was no significant difference between pyridine levels across the three different proxies. This implies that not only is pyridine not degrading over time like predicted, but it also does not interact with any of the non-monomer chemicals in the resin at all. For all the following discussion, the assumption is made that the measurement at time zero represents the true original concentration of pyridine added — meaning that there is assumed to be no degradation of pyridine on day 0.

Despite the lack of significant trends, a shelf life estimate can still be made. There exists a minimum shelf life that cannot be detected by the measurements performed due to error within the measurement process. A conservative estimate for the shelf life of the resin can be taken from

the maximum negative slope allowed within the 95% slope confidence interval of the resin proxy samples. This assumes a zero-order (linear) decomposition rate, which is independent of absolute concentration. Of all the data taken from the resin proxy samples, the most negative possible slope comes from Table 1, the resin proxy sample with measured ion 52m/z (aged before dilution). This sample has a maximum negative slope of -0.033 (normalized concentration per day). The shelf life for this resin has been defined as the time it takes for pyridine to decrease by 12% from its original level. This results in a conservative estimate of 3.636 day shelf life, or 3 days and 15 hours. This represents the minimum possible shelf life that could not be detected with the measurements taken. This is also close to empirical observations of resin quality over time — prints made from resin 3 or more days old have been observed to lose the soluble support functionality.

Thus, the data yields two possibilities. The first of these is that the resin has a shelf life greater than or equal to 3.636 days, and as such was not able to be detected through the error of the measurements. The other possibility is that pyridine does not degrade over time through the hypothesized pathways. If this is true, the resin's shelf life may be attributed to pyridine interactions with monomer (which was not included in the samples), or to another mechanism entirely.

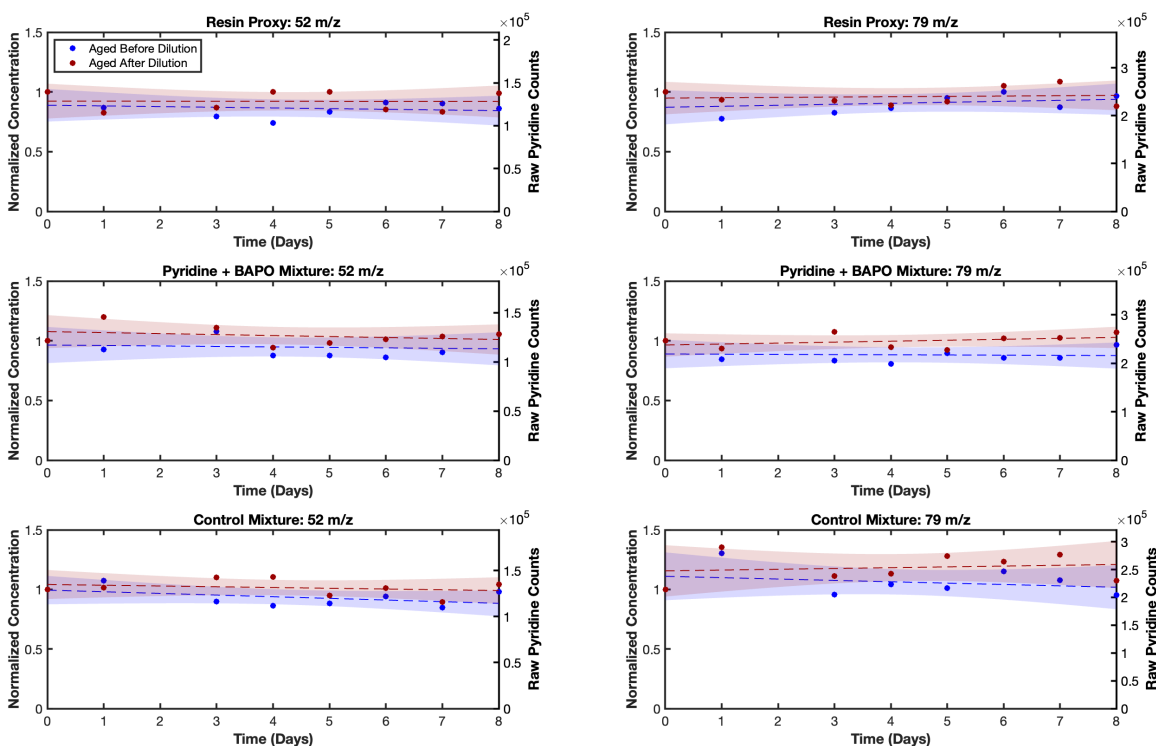


Figure 3: 95% confidence intervals and linear fits for each sample, each dilution method, and each measured pyridine ion. Across all samples, there was no significant change in pyridine levels over time. Left y-axis is pyridine counts normalized to counts measured at day 0, or the initial concentration. Right y-axis is counts of pyridine as measured directly from GC-MS.

Below are tables showing the slopes and 95% confidence on the slopes for the lines fitted to the data above.

Table 1: Normalized slope and 95% confidence interval on the slope for pyridine counts over time: 52 m/z.

52 m/z (Aged Before Dilution)	Control	Pyridine + BAPO	Full Proxy
Normalized Slope	-0.0139	-0.0040	-0.0057
95% Confidence	± 0.0238	± 0.0303	± 0.0273

52 m/z (Aged After Dilution)	Control	Pyridine + BAPO	Full Proxy
Normalized Slope	-0.0063	-0.0082	-0.0003
95% Confidence	± 0.0244	± 0.0276	± 0.0292

Table 2: Normalized slope and 95% confidence interval on the slope for pyridine counts over time: 79 m/z.

79 m/z (Aged Before Dilution)	Control	Pyridine + BAPO	Full Proxy
Normalized Slope	-0.0115	-0.0016	0.0083
95% Confidence	± 0.0402	± 0.0238	± 0.0286

79 m/z (Aged After Dilution)	Control	Pyridine + BAPO	Full Proxy
Normalized Slope	0.0067	0.0079	0.0028
95% Confidence	± 0.0431	± 0.0195	± 0.0272

As shown in the tables above, for every sample, the 95% confidence interval on the slope of the line fitted to the data includes 0. This means that there was no significant increase or decrease observed in the data over time. Because none of the linear fits to the data are significant, the data can effectively be modeled as $y = c$ (a flat line), where c is the significant fit parameter representing the average pyridine counts measured for each sample across all 9 days.

Uncertainty from GC-MS measurements is difficult to quantify without running a separate experiment, as it varies from instrument to instrument. Uncertainty from dilutions is also difficult to account for without having an internal standard because although the scale used for measurement is precise up to 0.01mg, evaporation of the mixtures during measurement can introduce variability. Ideally, two separate experiments would be run to determine the extents of the variability introduced by measurements. The variance in the particular GC-MS instrument being used would be determined by running multiple GC-MS measurements on the same exact sample. Error introduced by dilution can be characterized by preparing one sample, creating

multiple dilutions of it, and then running GC-MS on each dilution of the same sample. Between these two experiments, measurement uncertainty can be effectively quantified for the actual data.

Since it is unclear what the true uncertainty is, measurement uncertainty for this data is estimated to be 5% (the maximum relative standard deviation in response factor variation: from Agilent autosampler specification sheet [6]). This measurement uncertainty was then combined with the standard error on the mean pyridine counts for each sample. Mean pyridine counts for each sample and their respective combined uncertainties are represented in the tables below.

Table 3: Mean pyridine counts (non-normalized) for each sample and their respective total uncertainties for the pyridine ion 52 m/z.

52 m/z (Aged Before Dilution)	Control	Pyridine + BAPO	Full Proxy
Pyridine Counts	1.2118E+05	1.1501E+05	1.2056E+05
Total Uncertainty	1.6620E+04	1.6162E+04	1.6947E+04

52 m/z (Aged After Dilution)	Control	Pyridine + BAPO	Full Proxy
Pyridine Counts	1.3145E+05	1.2642E+05	1.2854E+05
Total Uncertainty	1.7341E+04	1.7076E+04	1.7969E+04

Table 4: Mean pyridine counts (non-normalized) for each sample and their respective total uncertainties for the pyridine ion 79 m/z.

79 m/z (Aged Before Dilution)	Control	Pyridine + BAPO	Full Proxy
Pyridine Counts	2.2739E+05	2.1726E+05	2.2633E+05
Total Uncertainty	3.4179E+04	2.9194E+04	3.1988E+04

79 m/z (Aged After Dilution)	Control	Pyridine + BAPO	Full Proxy
Pyridine Counts	2.5371E+05	2.4587E+05	2.3972E+05
Total Uncertainty	3.7227E+04	3.1539E+04	3.2589E+04

As shown by the tables, there is no significant difference in pyridine levels between any of the samples. There is also no significant difference between the two dilution methods, although some visual inspection indicates that the samples aged after dilution tend to have slightly higher pyridine levels than the samples aged before dilution. This could be related to evaporation effects that occur during sample preparation.

A noteworthy artifact which was not included in the analysis is the large presence of an unknown ion in all the samples. It is unlikely to be related to pyridine levels because its presence was relatively constant and it was present in the blank sample, but it was certainly unexpected.

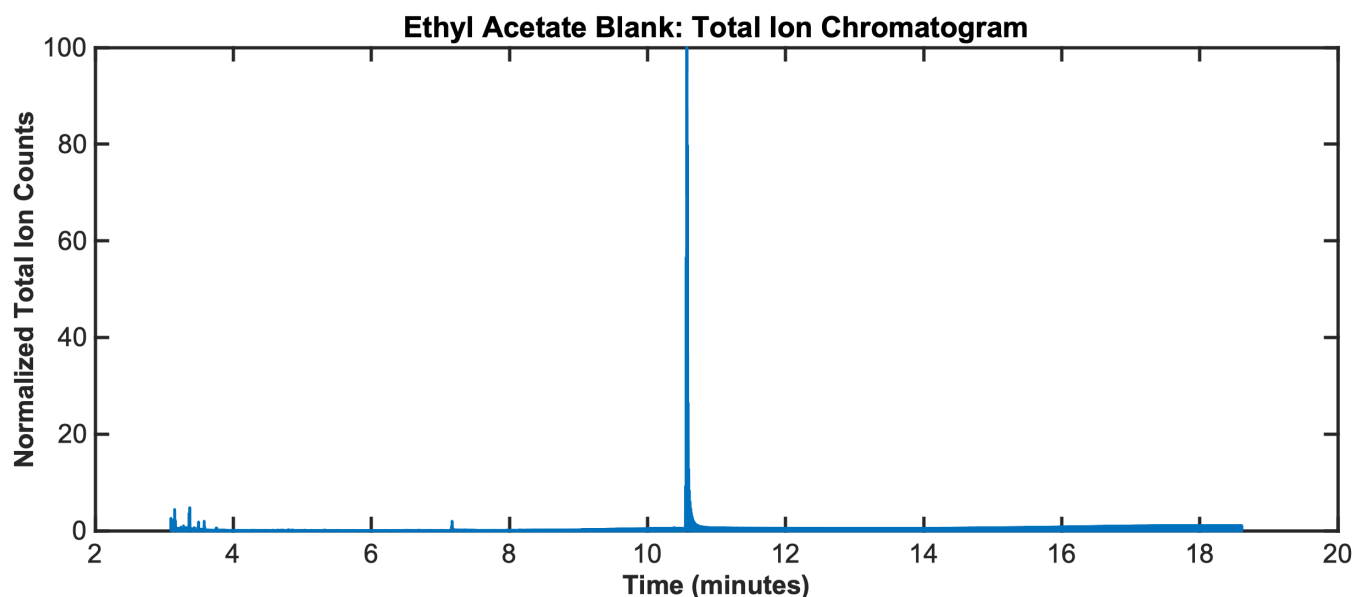


Figure 4: Total ion chromatogram for a blank sample containing just the GC-MS solvent. The large peak at roughly 10.5 minutes is most likely contamination of the instrument or column bleed. In many of the 52 m/z SIM graphs for the samples tested, there was a small trace corresponding to this peak.

Extracting the mass spectrum of this peak indicated that the main ions contributing to the peak have m/z values on the order of 200. This is certainly not pyridine, and is much more likely to be related to column bleed and/or contamination. However, the presence of a small peak at this time in the SIM graphs for the actual samples is potentially indicative of pyridine getting “stuck” to the column somehow, perhaps interacting with unknown ions in the instrument, and then eluting at a later time than normal. Ultimately, because there was no noteworthy change in the magnitude of this peak across all the samples, and because it does not normally represent a pyridine signal, this artifact was not incorporated into the analysis.

5. Conclusions

Investigation into the shelf life of the DLP 3D printing resin was performed through GC-MS measurements of pyridine over time. There was no significant decrease in pyridine levels over time for any of the samples tested — the control sample, the sample containing only pyridine and its suspected reactant, and the full proxy for the resin all had constant levels of pyridine throughout the time trial. Not only this, but pyridine levels were also not significantly different across any of the samples. This is another surprising result, because it indicates that pyridine in

fact reacts with none of the non-monomer ingredients in the resin. However, error in measurements can potentially conceal existing trends. Thus, the minimum possible shelf life which could not be detected by this method was found to be 3.636 days, relatively accurate to empirical observations of the resin's performance. While this is an exciting result, it is also entirely possible that pyridine does not decay over time, meaning that resin shelf life must be attributed to other mechanisms. Future work may need to be done to determine which of these is the most likely possibility.

In the meantime, researchers and engineers using this resin should still be mindful of the age and resultant quality of the resin. Even though pyridine depletion has not been conclusively shown to have an effect on the resin's shelf life, there are still many other potential internal interactions that this data does not account for, and it has not been proven that the resin is stable indefinitely.

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References

- [1] Diaco, Nicholas S., Carl J. Thrasher, Max M. Hughes, Kevin A. Zhou, Michael N. Durso, Saechow Yap, Robert J. Macfarlane, and A. John Hart. 2025. "Dual-Wavelength Vat Photopolymerization With Dissolvable, Recyclable Support Structures." *Advanced Materials Technologies* 10, no. 17 (June): e00650. <https://doi.org/10.1002/admt.202500650>
- [2] Marshall, David B., Edward M. Eyring, Frank Strohhusch, and Robert D. White. 1980. "Acid-Base Kinetics of Pyridine Studied with a Slow Spectrophotometric Indicator in Methanol." *Journal of the American Chemical Society* 102 (23): 7065-7067.
- [3] Hore, Nathan R., and Douglas K. Russell. "Radical Pathways in the Thermal Decomposition of Pyridine and Diazines: A Laser Pyrolysis and Semi-empirical Study." *Journal of the Chemical Society, Perkin Transactions 2* 1998, no. 2: 269-76. doi:10.1039/A706731C.
- [4] Behzadfar, A., Imani, M. & Farahmandghavi, F. Shelf-life of polyfurfuryl alcohol resin: an accelerated rheokinetics study. *Polym. Bull.* **76**, 5903–5918 (2019). <https://doi.org/10.1007/s00289-019-02692-4>

[5] Kalambet, Y., and Y. P. Kozmin. 2019. “Internal Standard Arithmetic Implemented as Relative Concentration/Relative Calibration.” *Journal of Chemometrics* 33, no. 6: 1-10. <https://doi.org/10.1002/cem.3106>

[6] Agilent Technologies. Agilent 7693A Automated Liquid Sampler — Data Sheet (5990-3526EN, December 6, 2021). Santa Clara, CA: Agilent Technologies, 2021.